NEUROFUSION AI | PRELIMINARY AI ANALYSIS REPORT

REPORT ID: REP-20261102-ABBA2E

Automated Multi-Modal AI Results • Stage 1: Pending Clinical Sign-Off

November 02, 2026 • 16:45 UTC

PATIENT INFORMATION	ATTENDING NEUROLOGIST	EXAMINATION DETAILS
ID: PAT-1083 **Name:** Eleanor Vance **Age/Gender:** 67 yrs / Female **Status:** Parkinson's Disease (Hoehn & Yahr Stage II)	**Name:** Dr. Clara Rodriguez, MD **License:** MD-NEURO-77491 **Dept:** Neurology & Movement Disorders **Hospital:** Metropolitan Neurological Inst.	**Scan Date:** 2026-11-02 **Report Date:** 2026-11-02 **Modality:** Voice **Device/Protocol:** NeuroFusionAI Protocol PARKINSON v2.0

PRIMARY AI PREDICTION: PARKINSON

Confidence: 89.5%

Target: Parkinson's Disease

Class Probability Distribution: Parkinson: 89.5% • Normal: 10.5%

AI Confidence Calibrated via Temperature Scaling

5. ELECTROENCEPHALOGRAPHY (EEG) NEUROPHYSIOLOGICAL ANALYSIS

Channels	Duration	Slowing Index	Dominant Band	Delta (0.5-4Hz)	Theta (4-8Hz)	Alpha (8-13Hz)	Beta (13-30Hz)
8 ch	12.5 s	1.48	**Theta**	35.0%	44.0%	14.0%	7.0%
EEG Interpretation:	Excess fronto-central theta power indicative of cortico-basal motor slowing.						

7. CLINICAL DECISION SUPPORT SYSTEM (CDSS) RECOMMENDATIONS

RISK LEVEL:
HIGH CLINICAL RISK

Urgency: Prompt Outpatient Specialist Review (Within 2-3 Weeks)

Evidence-Based Diagnostic Recommendations:
- Preliminary AI assessment indicates mild motor slowing.
- Recommend clinical neurological examination.

PATIENT LONGITUDINAL DIAGNOSTIC HISTORY

Date	Modality	Prediction	Confidence	Clinical Status
2026-11-02	Voice	Parkinson	89.5%	Verified Match
2026-11-02	Spiral	Parkinson	87.4%	Verified Match
2026-11-02	Voice	Parkinson	69.1%	Verified Match
2026-11-02	Voice	Parkinson	69.1%	Verified Match

PRELIMINARY AI SYSTEM OBSERVATIONS:
All required automated analytical pipelines (CNN/ViT, EEG feature extraction, XAI, and CDSS) have successfully executed. Final clinical diagnosis and treatment decisions await attending physician review.

CLINICAL APPROVAL STATUS
Preliminary AI analysis for decision support. Official physician certification pending.
Status: **AWAITING PHYSICIAN REVIEW & APPROVAL**

REGULATORY MEDICAL DISCLAIMER & CLINICAL INTENDED USE:
NeuroFusionAI is an investigational Artificial Intelligence Clinical Decision Support System (CDSS) designed to augment neurological assessment. All predictions, Grad-CAM attention heatmaps, and EEG feature extractions represent statistical patterns and MUST NOT be construed as a standalone or

confirmed medical diagnosis. Clinical treatment decisions must be made solely by a board-certified physician in conjunction with the patient's full medical history.